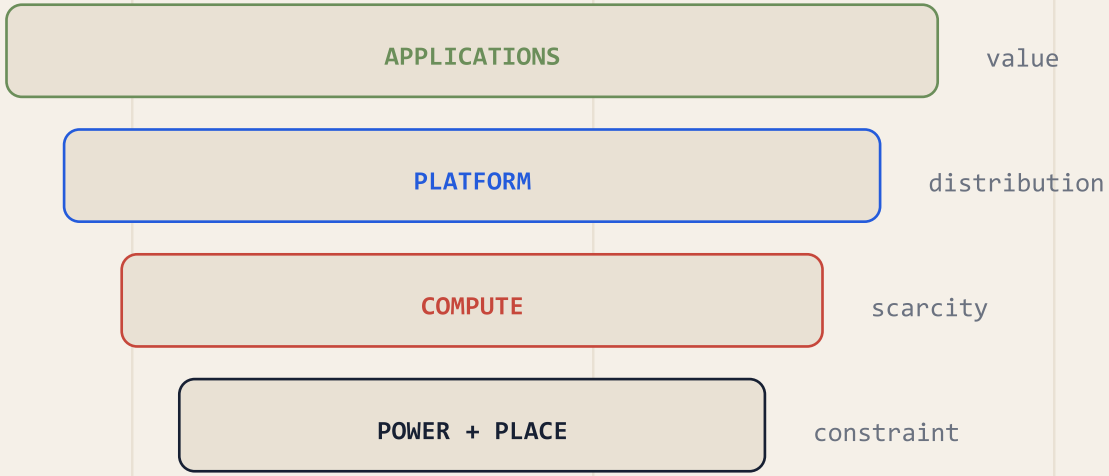
AI INFRASTRUCTURE ECONOMICS

Capital moves first. Constraints decide what ships.

A report on capacity, capital, and the physical stack.

A source-led editorial data study
for strategy and platform operators.

THE VALUE CHAIN IS PHYSICAL



PUBLIC DISCLOSURES / ILLUSTRATIVE SYSTEMS ANALYSIS

SOURCES: ALPHABET IR · MICROSOFT IR · SEE SOURCE NOTES IN PAGE PLAN

AI is not one market. It is a stack of dependencies.

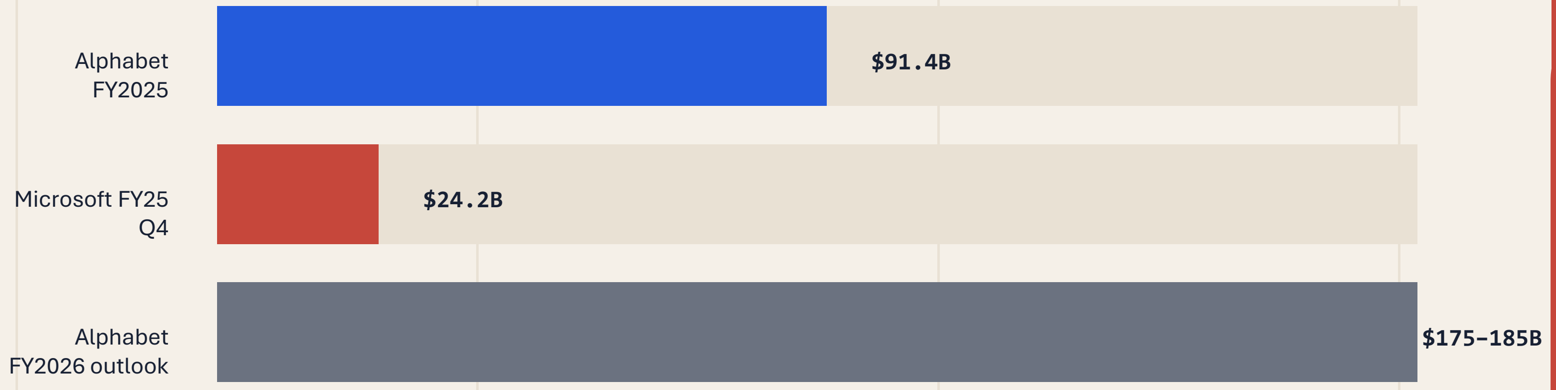
The margin pool can move upward; the bottleneck often stays physical.

- 07 / APPLICATIONS — workflow · distribution · willingness to pay
- 06 / MODELS — training · inference · evaluation
- 05 / SOFTWARE — orchestration · serving · observability
- 04 / SYSTEMS — accelerator · memory · network fabric
- 03 / DATA CENTERS — land · cooling · power delivery

dependency ↑

Capital is accelerating — but the periods are not interchangeable.

Reported figures are shown with company and fiscal-period labels, not as a false league table.



READ THIS AS

directional evidence
not a single-period ranking
different disclosure periods
different business mix

WHAT IS REPORTED

Large technical-infrastructure commitments are visible.

WHAT IT SUGGESTS

Capacity is becoming a strategic balance-sheet item.

WHAT IT DOES NOT PROVE

CapEx alone does not tell us who captures the margin.

SOURCES: ALPHABET 2025 Q4 EARNINGS CALL · MICROSOFT FY25 Q4 EARNINGS · FIGURES IN USD

Where the money lands changes the depreciation story.

One disclosed split makes the physical stack visible.

ALPHABET FY2025

$91.4B

technical infrastructure CapEx

APPROXIMATE 2025 COMPOSITION

SHORTER CYCLE

servers
faster refresh
utilization pressure

LONGER DURATION

data centers
networking
depreciation horizon

The stack has different clocks.

A server cycle can move faster than a building or grid connection.

SOURCE: ALPHABET 2025 Q4 EARNINGS CALL / APPROXIMATELY 60% SERVERS, 40% DATA CENTERS + NETWORKING

The constraint is not “compute.” It is the coordination of four physical systems.

Land, energy, networking, and servers arrive on different timelines.

LAND

permitted + buildable
lead time ≠ demand

ENERGY

predictable supply
lead time ≠ demand

NETWORK

fabric + interconnect
lead time ≠ demand

SERVERS

GPU + components
lead time ≠ demand

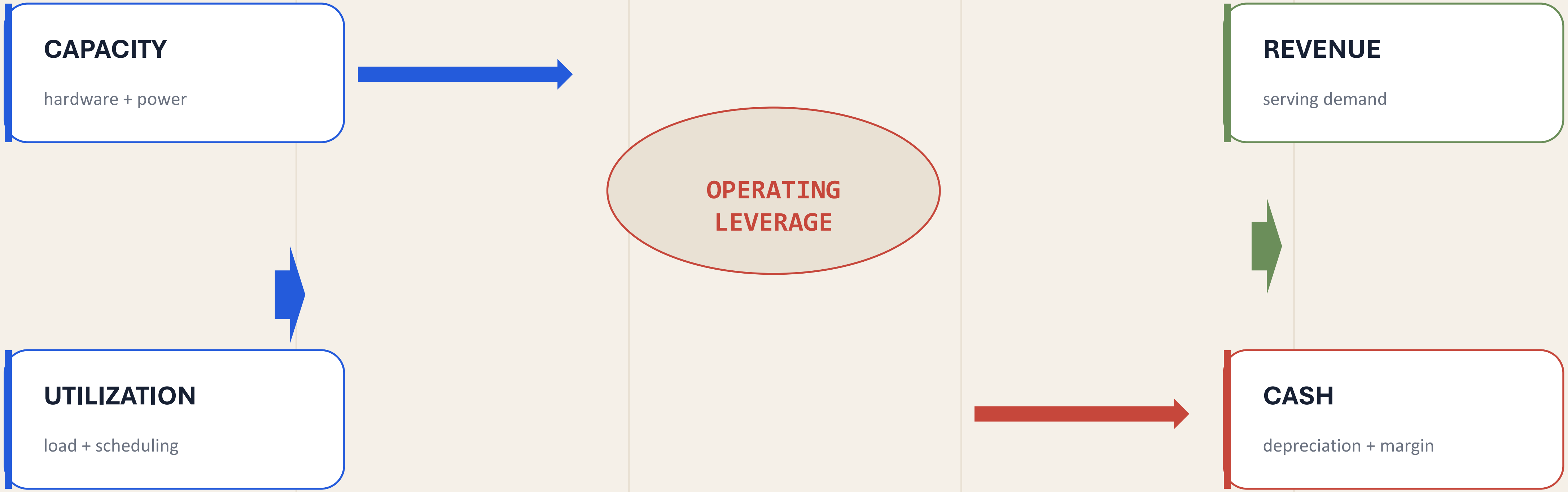
CAPACITY ONLY SHIPS WHEN THE SLOWEST LAYER ARRIVES.

This is why supply-chain risk belongs inside the AI strategy — not in a footnote after the model roadmap.

SOURCE: MICROSOFT FY2025 ANNUAL REPORT / DATA CENTER LAND, ENERGY, NETWORKING, SERVERS

Capacity becomes economics only through utilization.

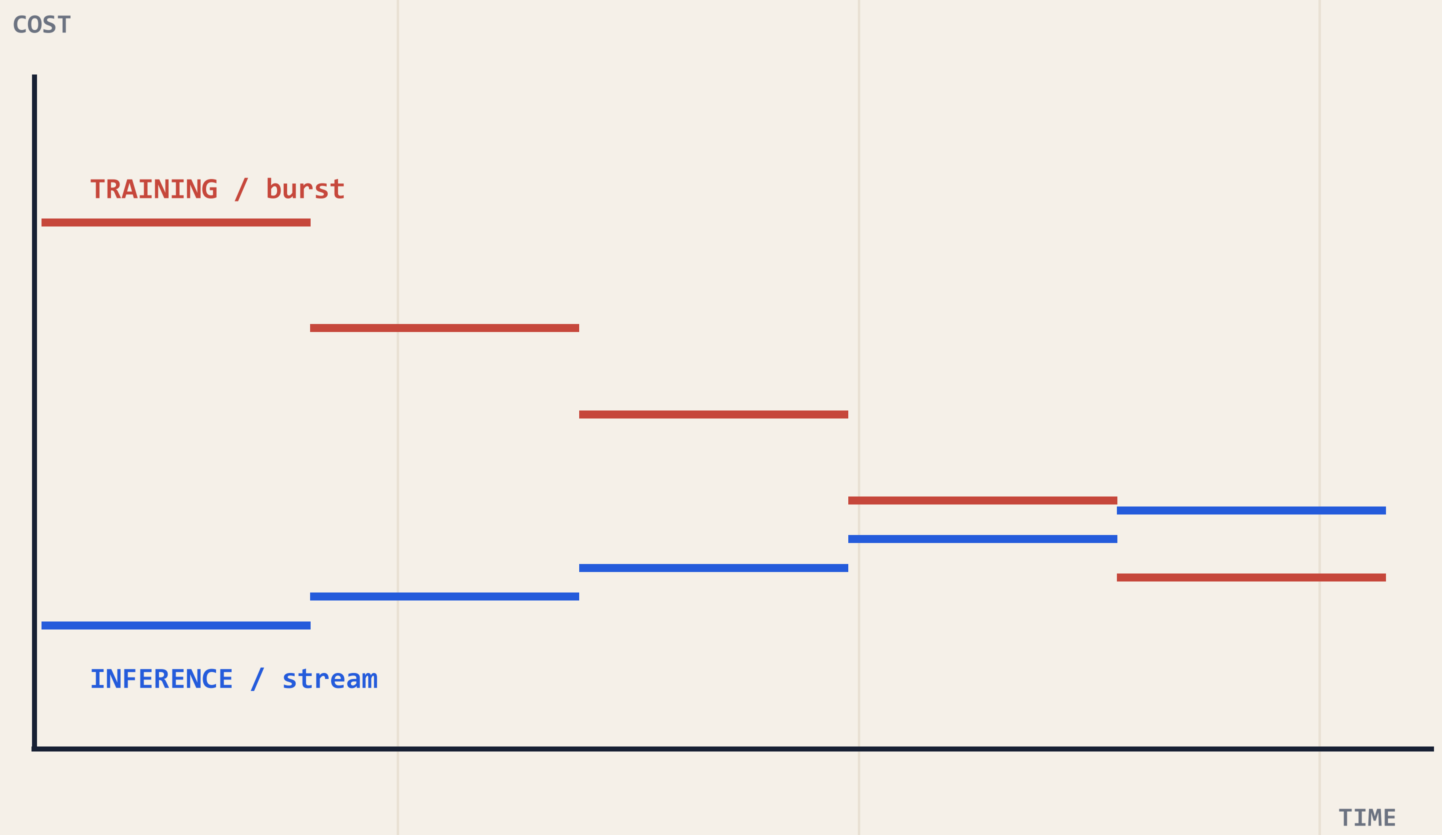
The flywheel is simple to draw — and difficult to keep turning.



NO UTILIZATION → NO LEVERAGE

Training is episodic. Inference is continuous.

Illustrative curves show why utilization and serving efficiency can matter more than a single benchmark.



OPERATOR QUESTION

Can the load be scheduled, shifted, or pooled?

ILLUSTRATIVE ONLY

Shape of curve is conceptual not a benchmark

The margin pool is a flow, not a single layer.

Compute creates the possibility; platforms and applications decide where value compounds.

POWER + PLACE

fixed assets

COMPUTE

scarce capacity

PLATFORM

orchestration

APPLICATION

workflow value

CAPITAL INTENSITY

low flexibility
high fixed cost

OPERATING LEVERAGE

shared capacity
better scheduling

CUSTOMER LOCK-IN

workflow fit
switching friction

The dangerous asset is capacity you cannot use.

A simple matrix makes the hidden risk visible: high lead time plus low utilization.

LEAD TIME

STRANDED
fixed cost
no flexibility

BOTTLENECK
demand exists
supply lags

UNDERUSE
capacity exists
demand is soft

HEALTHY
matched load
optionality

UTILIZATION →

WATCH ITEM

Lead time can outlive demand.

Treat capacity as an option until demand is proven.

There are three ways to buy capacity. None is free.

The right answer depends on demand certainty, capital appetite, and control requirements.

	BUY control	RENT optionality	BUILD differentiation
CAPITAL	high	variable	very high
SPEED	medium	fast	slow
CONTROL	high	shared	highest
RISK	demand	vendor	execution

OPERATOR TEST

How certain is demand? How unique is the workload?

Run infrastructure like an operating system, not a warehouse.

Five measures reveal whether the asset is earning its keep.

CAPACITY

what exists

UTILIZATION

what is used

LATENCY

what users feel

DEPRECIATION

what time costs

POWER

what scale requires

01

02

03

04

05

THE SCORECARD TURNS CAPEX INTO A FEEDBACK SYSTEM.

If one measure is missing, the operator is managing a blind spot — not a platform.

The winning infrastructure is not the biggest. It is the most legible.

Capital creates the option. Constraints define the clock. Control decides the return.

Capital creates the option.

Constraints define the clock.

Control decides the return.

THE OPERATOR’S QUESTION

Where is the bottleneck?
What can we flex?
What must we own?

MAKE THE STACK LEGIBLE.